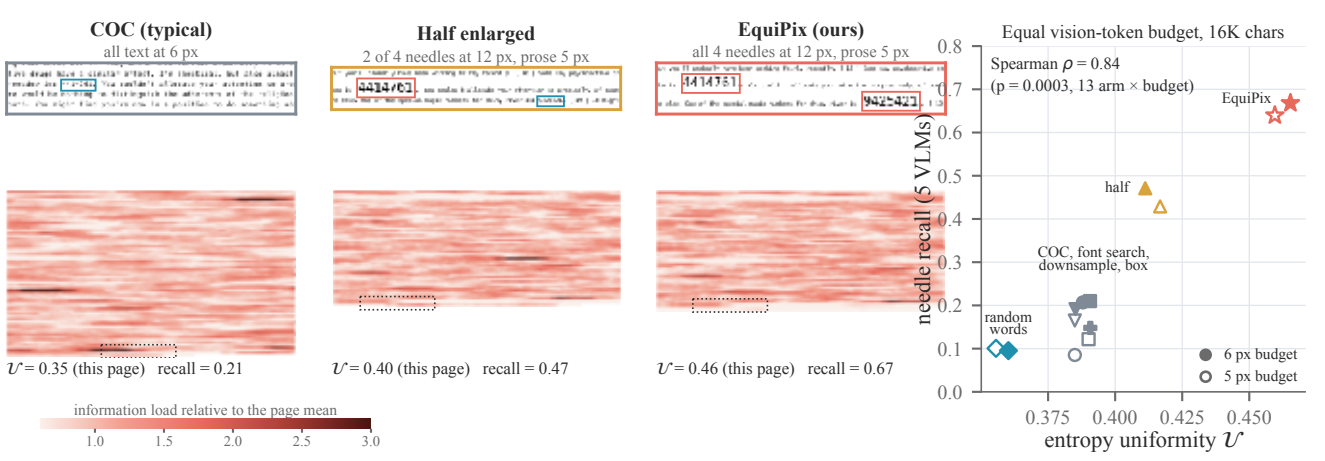

COC (typical)
all text at 6 px
Half enlarged
2 of 4 needles at 12 px, prose 5 px
EquiPix (ours)
all 4 needles at 12 px, prose 5 px
4414761
9425421
$\mathcal{U}^*$ = 0.35 (this page) recall = 0.21
$\mathcal{U}^*$ = 0.40 (this page) recall = 0.47
$\mathcal{U}^*$ = 0.46 (this page) recall = 0.67
information load relative to the page mean
1.0
1.5
2.0
2.5
3.0
Equal vision-token budget, 16K chars
Spearman ρ = 0.84
(p = 0.0003, 13 arm × budget)
EquiPix
half
COC, font search,
downsample, box
random
words
6 px budget
5 px budget
needle recall (5 VLMs)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.375
0.400
0.425
0.450
entropy uniformity $\mathcal{U}^*$